

Report Q2:

The counting system begins by isolating moving objects from the static scene using Mixture of Gaussian [MOG] background subtraction model. The technique models color history of every pixel using three distinct Gaussian distributions, learning what the background looks like over time. The model consists an adaptive learning rate α , to update the means and variance of these distributions, making sure that recent changes are implemented while older evidence becomes less relevant. The resulting mask is cleaned using morphological operations, and the remaining segments are filtered using aspect ratios in order to specifically target human shapes.

The system uses a particle filter to track individual movement across frames. This method represents each person as a cloud of 100 particles, estimating their states using a constant motion model. During the prediction phase, the filter projects these particles forward, adding random process noise to account for changes in speed or direction. To associate these predicted positions with the new raw detections, the TrackerManager uses the Hungarian algorithm. The particle filter then runs a correction phase where particles are weighted by their proximity to the actual measurement, then the resampled process, where successful particles are cloned and poor guesses are discarded.

The final counting mechanism make sure accuracy and eliminate false positives. A merge - detections.

- function first acts as a preprocessor, combining overlapping bounding boxes to fix instances where a person is split into multiple parts. The TrackerManager distinguishes transient noise from actual person.